TrackOps — Business Requirements Document (BRD)

Project Name: TrackOps | **Owner:** Owner / Admin | **Date:** 8 Oct 2026 | **Version:** 1.0 | **Status:** Draft

Linked docs: TrackOps PRD v1.3 · TrackOps API Documentation v1.0 · CHANGELOG.md (CR-001, CR-002) · Manual UAT sheet ("manual test trackops.xlsx")

*Items marked **TBC** are not stated anywhere in the project files and need the owner's input.*

1.1 One-Line Summary

TrackOps gives a small poster-stall owner a self-hosted, PC-based tool to track stock, sales, purchases and expenses with an accurate profit figure, replacing costly third-party tools and keeping all business data on the owner's own machine.

1.2 Business Problem

What's broken today? Existing business-management tools are expensive and keep business data on third-party servers. The owner (a small stall business selling posters in A4, A3, A5 and Polaroid sizes) has no single, affordable place to see stock levels, sales, purchases, expenses and profit together.

Why does it matter now? This is a personal project meant to support the owner's day-to-day business tracking directly. A one-month delivery window was set, with v1.0 targeted for 12 Nov 2026. Without it, stock and profit decisions rely on tools the owner either cannot afford or does not trust with their data.

1.3 Current State (As-Is Snapshot)

Aspect	Current Reality
Process/System in use	Commercial business-management tools are considered too expensive; the owner's actual day-to-day method (paper, spreadsheet, none) is **TBC**
Time/Cost involved	Subscription cost of commercial tools and weekly manual effort: **TBC** (no figures recorded)
Who is impacted	The owner, who is currently the only user

Aspect	Current Reality
Known pain points	High tool cost; business data stored on third-party servers; no unified view across inventory, sales, purchases and expenses; no reliable profit calculation that accounts for cost of goods sold

1.4 Business Objectives

#	Objective	Success Metric	Target
1	Manage inventory	Products can be added, edited, deleted; stock can be adjusted with a reason; every stock change is logged	Working by 12 Nov 2026
2	Record sales accurately	A saved sale reduces stock and appears as money in; stock cannot go negative	Working by 12 Nov 2026
3	Record purchases and expenses	A saved purchase increases stock and appears as money out; manual expenses and other income can be logged	Working by 12 Nov 2026
4	Know true profit	Net Profit = Sales + Other Income − COGS − Operating Expenses, with stock purchases kept out of expenses	Figures match on-screen and in exports
5	Generate reports	Profit & Loss, registers and period comparison available, exportable to PDF and Excel	Working by 12 Nov 2026
6	Own the data	All data held in a PostgreSQL database on the owner's PC; no third-party servers; works offline	Zero external data services

Delivery status (from project files): Objectives 1–6 are implemented. Objective 5 is partly delivered: the P&L, registers and PDF/Excel exports are live; the visual-charts Reports page was deliberately postponed to the next version (CR-001 amendment). A manual UAT sheet lists 24 test cases across Inventory, Purchases, Sales and Finances; all are marked **pass**.

1.5 Scope

In Scope:

- Inventory: product catalogue (name, size, stock, cost price, selling price), stock adjustment with reason, product deletion that preserves history, stock movement log.
- Sales: billing counter, Cash/UPI payment, automatic stock reduction, receipt, edit and delete of saved sales with stock restored.
- Purchases: stock purchase vouchers, automatic stock increase, latest-cost pricing, edit and delete with stock safeguards.
- Finances: manual expenses and other income, Expense Register and Income Register, Profit & Loss with previous-period comparison, PDF and Excel export.
- Local PostgreSQL database and Express API on the owner's PC; responsive React web interface.
- Indian Rupee (₹) with Indian digit grouping; all date boundaries in Asia/Kolkata.

Out of Scope:

- Third-party integrations (future versions)
- Login, staff accounts and multi-user roles
- Credit sales and partial payments (every sale is paid in full)
- GST and tax handling
- Languages other than English
- Native Android or iOS apps
- Customer and supplier management; payroll; multi-location stock
- Visual charts Reports page (code parked in `src/parked/ReportsPage.tsx` for the next version)
- Home page (removed under CR-001)

Assumptions:

- Single admin user; the app is used only while the owner's PC is running (low, roughly weekly usage).
- Phone and PC are on the same private network; remote access has been tested over Tailscale.
- A product's cost price follows the latest purchase cost (not a weighted average) — assumed, awaiting owner confirmation.
- Purchases are money out but not a P&L expense; they reach profit only as COGS when sold — assumed, awaiting owner confirmation.

- Questions that were open in PRD v1.3 and have since been settled by the build: the technology stack (React, TypeScript, Vite, Tailwind, Express, PostgreSQL), and whether saved sales can be edited or cancelled (yes, with stock restored).
- **Document drift to note:** PRD v1.3 still describes a Home page and a Reports section in its user flow and has no feature list; the built product removed Home and parked the charts page. The PRD should be updated to match.

1.6 Stakeholders

Name/Role	Interest/Stake	Approval Authority?
Owner / Admin	Sole business owner and only user; needs accurate stock, sales and profit data under their own control	Yes

1.7 Business Requirements

ID	Requirement	Priority	Linked Objective
BR-01	The business must be able to maintain a product catalogue with size, cost price, selling price and stock quantity.	Must	1
BR-02	Every stock change (opening stock, sale, purchase, adjustment) must be recorded with before/after balance so stock can be audited.	Must	1
BR-03	Stock must never go negative; sales that exceed available stock must be refused.	Must	1, 2
BR-04	Deleting a product must not erase past sales, purchases or stock history.	Must	1
BR-05	The business must be able to record a sale (Cash or UPI) that reduces stock and is counted as money in.	Must	2
BR-06	Cost at the time of sale must be locked so profit stays correct when prices later change.	Must	2, 4

ID	Requirement	Priority	Linked Objective
BR-07	The business must be able to record stock purchases that increase stock and are counted as money out.	Must	3
BR-08	The business must be able to record manual operating expenses and other income by category, date and payment method.	Must	3
BR-09	Saved sales and purchases must be correctable (edit or delete) without breaking stock or financial totals.	Should	2, 3
BR-10	Profit must be calculated as Sales + Other Income − COGS − Operating Expenses; purchases must not be treated as operating expenses.	Must	4
BR-11	The business must be able to view Money In, Money Out and Net Profit for any date range, with comparison to the previous period of equal length.	Must	4, 5
BR-12	Financial statements and registers must be exportable to PDF and Excel with figures matching the screen.	Must	5
BR-13	All business data must stay in a database on the owner's PC, with no dependency on third-party servers.	Must	6
BR-14	The application must work on both phone and desktop browsers, using ₹ and Asia/Kolkata dates.	Must	7
BR-15	Visual charts and trend reports.	Could (next version)	5
BR-16	Backup and restore of the PostgreSQL database.	Should (not yet defined)	6

1.8 Cost–Benefit Snapshot

	Estimate
Rough cost/effort	About one month of the owner's own build effort; no software licence cost (open-source stack) and no hosting fee (runs on the owner's existing PC). Monetary value of effort: **TBC**
Expected savings/benefit	Avoids subscription fees of commercial tools (amount **TBC**); keeps data private; gives a reliable profit figure based on cost of goods sold
Payback period	**TBC** — depends on the subscription cost avoided
Risk of doing nothing	Continued reliance on costly tools or manual tracking, business data held by third parties, and no trustworthy profit view

1.9 Constraints

- Delivery within one month (v1.0 target 12 Nov 2026).
- Must run on the owner's PC with a local PostgreSQL database; the app is available only while that PC is on.
- Single-user, no login in v1; access relies on a private network.
- Currency ₹ only; English only; no GST.
- Web application only (no native apps).

1.10 Risks at a Glance

Risk	Impact	Likelihood
PC is off, unreachable or fails, so the app is unavailable or data is lost	High	Medium
No database backup process is defined for PostgreSQL	High	Medium
No login while the server listens on all network interfaces with open cross-origin access; anyone on the same network could read or change data	High	Medium

Risk	Impact	Likelihood
Requirements documents (PRD v1.3) are out of date against the built product	Medium	High
Reports charts postponed, so visual reporting is not available in v1	Low	Certain
Tight one-month timeline for a responsive web app	Medium	Low (core build is complete)

1.11 Dependencies

- Owner's PC running, with PostgreSQL installed and the `DATABASE_URL` configured.
- Phone and PC on the same network (or Tailscale) for mobile access.
- Node.js environment to run the API (port 4000) and web interface (port 3000).
- Final owner approval of the open assumptions in section 1.5.

1.12 Sign-off

Name	Role	Decision	Date
Owner	Owner / Admin	complete	20/10/2026